

Mir Mohammad Yousuf

Quantum Software Engineering Researcher & Educator

Reliable Quantum Computing | Noise-Aware Testing | Quantum Machine Learning

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ACADEMIC PROFILE

Researcher and educator combining doctoral research in Quantum Software Engineering at the National Institute of Technology Srinagar with **over four years of university teaching experience** as an Assistant Professor at the University of Kashmir and Lovely Professional University. Research focuses on reliable quantum computing, interpretable defect analysis, and noise-aware testing, with recent first-author publications in *Automated Software Engineering* and *Neurocomputing*. Contributions span large-scale empirical studies and self-calibrating testing evaluated on IBM Quantum hardware, alongside co-authored work on adaptive error mitigation for **Quantum Machine Learning and quantum reinforcement learning**.

Teaching and mentorship span undergraduate and postgraduate instruction in software engineering, programming, data structures, algorithms, computer networks, and research methodology. Guided **student research projects and theses**, contributed to curriculum development, and mentored student teams; continues to support coursework and laboratory instruction at NIT Srinagar. Brings experimental design, statistical analysis, IBM Qiskit, and industry software quality-assurance experience to research, practical instruction, and student research supervision.

RESEARCH INTERESTS

- Reliable quantum computing; Quantum Software Engineering; noise-aware testing and verification.
- Quantum Machine Learning; robustness, reproducibility, and testing of hybrid quantum-classical learning workflows.
- Empirical evaluation of NISQ software; noise sensitivity, backend-dependent behaviour, and software quality.
- AI/ML for Software Engineering and Software Quality; interpretable defect analytics; repository mining and software maintenance.

EDUCATION

Doctor of Philosophy (Ph.D.) in Information Technology

Oct. 2022 – Present

National Institute of Technology Srinagar, India

Research area: Quantum Software Engineering. **Thesis:** *Quantum Software Engineering: Bridging Theory and Practice*.

Supervisor: Dr. Shabir Ahmad Sofi. **Focus:** Quantum-software defects, quality attributes, and noise-aware testing.

Status: Minimum requirements completed for thesis submission.

M.Tech. in Computer Science and Technology

2016 – 2018

Central University of Punjab, Bathinda, India

Result: Distinction. **Thesis:** *Analysis of the Change of Bugginess and Adaptiveness of Python Software Systems*.

B.Tech. in Computer Science and Engineering

2011 – 2015

University of Kashmir, Srinagar, India

Result: Distinction; ranked fourth among the university's Computer Science and Engineering students.

ACADEMIC EXPERIENCE

Assistant Professor

Jun. 2021 – Oct. 2022

University of Kashmir

Taught undergraduate and postgraduate courses in programming, algorithms, Python, software engineering, data structures, and research methodology. Contributed to curriculum development for bachelor's and master's programmes and supported project coordination and student mentoring.

Taught software engineering, computer networks and laboratories, C and Python programming, data structures, and research methods. Guided student research projects and theses; coordinated academic activities and mentored students.

RESEARCH EXPERIENCE

Research Fellow & Teaching Assistant

Oct. 2022 – Present

Department of Information Technology, National Institute of Technology Srinagar

- Systematically synthesized Quantum Software Engineering methods and lifecycle practices and developed a structured taxonomy of challenges in the emerging field.
- Designed an interpretable rule-based framework for quantum-software bug classification; evaluated it on 12,910 issues from 36 Qiskit repositories and 4,984 manually annotated cases, demonstrating strong agreement with human judgment and advantages over standard machine-learning baselines on difficult fine-grained categories.
- Extended the defect-research programme through cross-ecosystem classification and a longitudinal empirical analysis of 32,296 verified bug reports from 123 open-source repositories.
- Developed a noise-aware testing framework based on self-baseline calibration, deriving program- and backend-specific correctness envelopes and evaluating them across multiple noise regimes, hardware snapshots, and a live IBM Quantum system.
- Contributed to coursework and laboratory instruction alongside doctoral research.

RECENT PUBLICATIONS

A. Published and Accepted

1. **M. M. Yousuf** and S. A. Sofi, “Bug Classification in Quantum Software: A Rule-Based Framework and Its Evaluation,” *Automated Software Engineering*, Springer, vol. 33, no. 1, art. 39, 2026.
2. **M. M. Yousuf** and S. A. Sofi, “A Systematic Exploration of Quantum Software Engineering in the NISQ Era: Methods, Lifecycle Practices, and a Taxonomy of Challenges,” *Neurocomputing*, Elsevier, art. 132809, 2026.
3. **M. M. Yousuf**, “Cross-Ecosystem Bug Classification in Quantum Software,” *International Conference on Communication and Computational Technologies (ICCMC-183)*. [Accepted]
4. B. Majid, S. A. Sofi, and **M. M. Yousuf**, “APGEM: Adaptive Policy-Guided Error Mitigation for Quantum Reinforcement Learning on a Real-World CVRP Case Study.” Accepted at the *2026 6th International Multi Conference on Artificial Intelligence Technology (MCAIT)*. [Accepted]

B. Under Review and Submitted

1. **M. M. Yousuf** and S. A. Sofi, “Characterizing Bugs and Quality Attributes in Quantum Software: A Large-Scale Empirical Study,” arXiv preprint arXiv:2512.24656, 2025. Under review at *Quantum Information Processing*.
2. **M. M. Yousuf** and S. A. Sofi, “Noise-Aware Testing for NISQ Quantum Software: A Framework and Empirical Characterization of Testing Boundaries.” Under review at *ACM Transactions on Software Engineering and Methodology (TOSEM)*.
3. **M. M. Yousuf** and S. A. Sofi, “Noise-Awareness in Quantum Software Testing: An Empirical Landscape Analysis.” Under review at *Automated Software Engineering*.
4. B. Majid, S. A. Sofi, and **M. M. Yousuf**, “Adaptive Policy-Guided Error Mitigation for Noise-Resilient Quantum Reinforcement Learning on the Capacitated Vehicle Routing Problem.” Submitted to *Cities*.
5. B. Majid, S. A. Sofi, and **M. M. Yousuf**, “Adaptive Policy-Guided Error Mitigation (APGEM): A Unified Learning-Based Framework for Noise Resilience in Quantum Machine Learning.” Submitted to *Engineering Applications of Artificial Intelligence*.

SELECTED RESEARCH CONTRIBUTIONS

Interpretable Quantum-Software Bug Classification

Created a rule-based framework for classifying quantum-software defects, providing interpretable categories for defect analysis. Evaluation against manually annotated cases demonstrated strong agreement with human judgment and advantages over standard machine-learning baselines on difficult fine-grained categories.

Quantum Software Engineering Methods & NISQ Challenges

Synthesized methods and lifecycle practices across a fragmented research field and developed a structured taxonomy of NISQ-era challenges, organizing the evidence around the engineering of quantum software.

Longitudinal Quantum-Software Quality

Characterized bugs and quality attributes over time through a large-scale study of verified reports from open-source quantum-software repositories, extending the research from defect classification to longitudinal software-quality analysis.

Self-Calibrating Noise-Aware Testing

Developed a methodology that replaces fixed correctness thresholds with program- and backend-specific correctness envelopes derived through self-baseline calibration. Evaluated it across noise regimes, hardware snapshots, and a live IBM Quantum system. Complementary landscape analysis characterized how noise awareness is addressed in quantum-software testing.

Cross-Ecosystem Quantum Bug Classification

Extended Qiskit-focused defect classification across quantum-software ecosystems to examine the broader applicability and transferability of bug categories beyond a single ecosystem.

Reliable Quantum Machine Learning & Reinforcement Learning

Co-authored research on Adaptive Policy-Guided Error Mitigation (APGEM), a learning-based approach to noise resilience in QML, and its application to quantum reinforcement learning for the Capacitated Vehicle Routing Problem, including a real-world case study.

TEACHING, SUPERVISION & CURRICULUM DEVELOPMENT

- **Teaching areas:** Software Engineering, Software Quality Assurance, Computer Networks, C and Python Programming, Data Structures, Algorithms, and Research Methodology; undergraduate and postgraduate instruction.
- **Research supervision:** Guided student research projects and theses at Lovely Professional University.
- **Curriculum development:** Contributed to curriculum development for undergraduate and postgraduate programmes at the University of Kashmir.
- **Laboratory instruction and mentoring:** Delivered programming and computer-network laboratory instruction and research-methods teaching; supported coursework and laboratories at NIT Srinagar and mentored a seven-member Smart India Hackathon 2019 team.

INDUSTRY & SOFTWARE ENGINEERING EXPERIENCE

Automation Engineer

Dec. 2017 – May 2018

Central University of Punjab

Automated student-management and result-system modules; implemented PHP/MySQL components and performed black-box quality assurance.

Quality Assurance and Control Engineer

Nov. 2015 – Mar. 2016

BQE Software Inc.

Tested BillQuick, WebSuite, ArchiOffice, BQE Online, and ArchiOffice Online using black-box and white-box techniques across the Software Testing Life Cycle.

Software Developer and Trainer

Dec. 2014 – Jan. 2016

IAC Software Solutions

Developed web applications with HTML, CSS, PHP, and SQL and delivered training in C, C++, Java, and database design.

INVITED TALKS & ACADEMIC ENGAGEMENT

- **Invited Speaker, NIT Srinagar, 2026** – invited session on “Quantum AI.”
- **Invited Expert, AI Summit 2026** – Skill Development Centre, Sher-i-Kashmir Institute of Medical Sciences (SKIMS), Srinagar; official pre-summit event on AI in healthcare (2026).
- **Invited Speaker, National Conference on Patient Safety 2025** – “Artificial Intelligence: Safeguarding Patients in the Era of Digital Healthcare,” SKICC, Srinagar (2025).
- **Expert Resource Person, Python for Artificial Intelligence Workshop** – Department of Information Technology, NIT Srinagar, 6–10 Oct. 2025.
- **Invited Expert, Codevolution 2025** – ICT Department, Green Valley Educational Institute, Srinagar (2025).

ACADEMIC SERVICE

- Registration and website committee, Indian Science Congress hosted by Lovely Professional University (2019); received a Certificate of Appreciation.
- Mentor, Smart India Hackathon 2019, Ministry of Water Resources problem statement.
- Academic and project coordination, student mentoring, and organisation of *Convergence-17*, Central University of Punjab (2017).

AWARDS & DISTINCTIONS

- **Best Paper Award**, International Conference on Advances in Science and Technology (ICAST), 2018.
- **Second rank nationally**, Computer Science Learning Test (CSLT), 2020.
- **Fourth rank**, B.Tech. Computer Science and Engineering, University of Kashmir, 2015.

TECHNICAL EXPERTISE

Quantum computing & experimentation: IBM Qiskit, QSim, noise-aware testing, self-baseline calibration, and experimental evaluation across noise regimes and IBM Quantum hardware.

Empirical research methods: Repository mining, manual bug annotation, longitudinal analysis, experimental design, hypothesis testing, correlation, and regression.

Machine learning & data analysis: Applied machine learning, statistical analysis, and data analysis.

Software engineering & testing: Software Testing Life Cycle, black-box testing, white-box testing, software quality assurance, and Selenium.

Programming: Python, C/C++, Java, SQL, PHP, JavaScript, and HTML/CSS.

Research & development tools: L^AT_EX, Linux, Android Studio, Git, Jupyter Notebook, and MATLAB.

CERTIFICATIONS AND PROFESSIONAL DEVELOPMENT

Quantum Computing

- Practical Quantum Computing with IBM Qiskit for Beginners – Packt (Oct. 2025). Credential ID: 5YEHGHNZRTXL.
- Quantum Computing for Everyone: An Introduction – Fractal (Oct. 2025). Credential ID: TVMUKRJZFN6W.
- AI-Assisted Software Engineering for Quantum Computing and PQC (AI-SEQ) – QuLearnLabs (Oct. 2025).
- Quantum Computing: Theory to Physical Implementation – Indian Institute of Science (IISc) (May 2025).
- Quantum Computing using Indigenous Quantum Simulator (QSim) – Centre for Development of Advanced Computing (C-DAC) (Jul. 2023).

Software Engineering and Testing

- Software Engineering: Implementation and Testing – Hong Kong University of Science and Technology (Oct. 2024). Credential ID: 4JHZMJRC6SCX.
- Introduction to Software Engineering (with Honors) – IBM (Aug. 2024). Credential ID: AVHGTHQIKMZL.
- Software Engineering Essentials – IBM (Aug. 2024). Credential ID: AVHGTHQIKMZL.
- Web and Mobile Testing with Selenium – University of Minnesota (Aug. 2024). Credential ID: 70HFKDW4TIB0.
- Foundations of Software Testing and Validation – University of Leeds (Jun. 2024). Credential ID: JSKJL49QZQ48.

Programming and Data Science

- Python for Data Science, AI & Development – IBM (Oct. 2024). Credential ID: LPJOSJRJIGJV.
- Programming in Python – Meta; completed Oct. 2024. Credential ID: B3A0PXAP0PRX.
- Python 101 for Data Science – Cognitive Class (Mar. 2019); successfully completed.

Networking, Languages, and Development

- CompTIA Network+ – CompTIA (Dec. 2019).
- English Proficiency Test – EF SET.
- Android Development – IAC Software Solutions (Jul. 2015).
- Java – IAC Software Solutions (Jul. 2015).
- CCNA – University of Kashmir (Feb. 2015); Grade A.

LANGUAGES

English (fluent) • Urdu (fluent) • Kashmiri (native) • Arabic (reading and writing)

PUBLICATIONS: EARLIER WORK

A. Journals

1. **M. M. Yousuf** and M. Rashid, "Analysis of the Change in Bugginess and Adaptiveness of Bug Tracking Trac Systems for Python Softwares," *Multimedia Tools and Applications*, Springer Nature, 2022. DOI: 10.1007/s11042-022-13246-8.
2. **M. M. Yousuf**, B. Majid, M. Rashid, and U. I. Wani, "On the Statistical Relationships Between the Coding Standards and the Reported Bugs of the Python Software Systems," *Think India*, vol. 22, no. 3, pp. 7718–7727, Jul. 2019.
3. U. I. Wani, **M. M. Yousuf**, and M. Rashid, "IoT Architecture, Challenges and Future Directions – A Mirror Review," *Think India*, vol. 22, no. 3, pp. 8402–8410, Jul. 2019.
4. U. I. Wani, G. Singh, **M. M. Yousuf**, M. Rashid, and B. Talwar, "A Brief Survey on Various Security Attacks in IoT," *Think India Journal*, vol. 22, no. 3, pp. 8392–8401, Jul. 2019.
5. M. Rashid, A. Kaur, **M. M. Yousuf**, and R. Puri, "Comparative Analysis of Big Data Projects for Solving Computational Challenges in Data," *Think India Journal*, vol. 22, no. 3, pp. 8090–8096, Jul. 2019.
6. **M. M. Yousuf**, R. Kumar, M. Singh, R. Yadav, and Isha, "Emotion Analysis with Deep Learning and Machine Learning Technologies," *Think India Journal*, vol. 22, no. 3, pp. 7728–7737, Jul. 2019.
7. **M. M. Yousuf**, S. Kar, M. Rashid, and B. Majid, "A Comprehensive Review on the Different Types of the Attacks and Defense in Deep Learning," *Think India Journal*, vol. 22, no. 3, pp. 7738–7748, Jul. 2019.
8. **M. M. Yousuf**, S. Kumar, M. Rashid, and B. Majid, "Digital Data Forgetting: A Machine Learning Approach," *Think India Journal*, vol. 22, no. 30, pp. 318–324, Dec. 2019.
9. M. Rashid, **M. M. Yousuf**, A. Kaur, R. Puri, and V. Goyal, "Comparative Studies on Text Extraction from Real Time Images," *Think India Journal*, vol. 22, no. 16, pp. 2138–2142, Aug. 2019.
10. A. Ahmed, A. Kumar, and **M. M. Yousuf**, "Using AZ Software Development Model to Improve the Quality of a Software Application," *JETIR*, vol. 6, no. 1, pp. 11–18, Jan. 2019.
11. **M. M. Yousuf**, "Is It Safe to Uplift a Patch? An Empirical Study on Trac," *International Journal of Advance Research in Science and Engineering*, 2018.
12. S. Soni, J. Singla, and **M. M. Yousuf**, "Big Data: Frameworks and Challenges," *International Journal of Emerging Technologies and Innovative Research (JETIR)*, vol. 5, no. 10, pp. 535–541, Oct. 2018.
13. V. K. Pandey, J. Singla, and **M. M. Yousuf**, "Mobile Cloud Computing: A New Opportunity for the Rapidly Changing World," *JETIR*, vol. 5, no. 10, pp. 566–569, Oct. 2018.
14. K. Rana and **M. M. Yousuf**, "Comparative Analysis of Testing Tools," *JETIR*, vol. 5, no. 10, pp. 585–591, Oct. 2018.
15. N. Saqib and **M. M. Yousuf**, "A Review of Mitigating Cost Concerns to Enhance Maintainability in SDLC," *JETIR*, vol. 5, no. 11, pp. 162–169, Nov. 2018.
16. N. Saqib, **M. M. Yousuf**, K. Rana, and Kavita, "A Review on the Internet of Things for Healthcare," *JETIR*, vol. 5, no. 11, pp. 244–253, Nov. 2018.
17. Chandini, J. Singla, and **M. M. Yousuf**, "Denial of Service Attack in Internet of Things," *JETIR*, vol. 5, no. 12, pp. 920–928, Dec. 2018.
18. A. A. Gadoo, **M. M. Yousuf**, M. Rashid, and S. Khare, "A Survey on Source Camera Identification Using Image Features," *JETIR*, vol. 5, no. 12, pp. 989–999, Dec. 2018.
19. Divya K. and **M. M. Yousuf**, "A Review on Image Processing and Image Segmentation," *IJRAR*, vol. 5, no. 4, pp. 310–321, Nov. 2018.
20. A. Paliwal, M. Dhindsa, **M. M. Yousuf**, and Roopam, "Malware Detection and Prevention Based on Machine Learning Algorithms," *IJRAR*, vol. 5, no. 4, pp. 205–214, Dec. 2018.
21. Sk. Md. Alfayeed, **M. M. Yousuf**, and M. Dhindsa, "An Overview and Applications of Artificial Intelligence and Robotics," *IJRAR*, vol. 5, no. 4, pp. 783–787, Nov. 2018.
22. G. S. Reddy, **M. M. Yousuf**, and S. M. Lakshmi, "Review Based on Detection System Using Web Content Mining in Cyberbullying," *IJRAR*, vol. 5, no. 4, pp. 1094–1098, Dec. 2018.
23. **M. M. Yousuf**, "A Comparative Study on Agile Software Development Technologies," *IJRAR*, vol. 5, no. 4, pp. 1112–1119, Dec. 2018.
24. Y. Kumar and **M. M. Yousuf**, "A Review on Wireless Sensor Network," *IJRAR*, vol. 5, no. 4, pp. 442–451, Dec. 2018.
25. A. Kumar, J. Singla, and **M. M. Yousuf**, "An Overview of Windows Update, Security Patch and Issues," *IJRAR*, vol. 5, no. 4, pp. 1077–1084, Dec. 2018.
26. M. Rashid and **M. M. Yousuf**, "Crop Disease Detection Using Image Processing Techniques – A Systematic Review," *Journal of Gujarat Research Society*. <http://www.gujaratresearchsociety.in/index.php/JGRS/article/view/1324/1093>
27. S. Khare, **M. M. Yousuf**, A. A. Gadoo, and A. K. Shukla, "Virtualization and Security in Cloud Computing," *JETIR*.

B. Conferences

1. **M. M. Yousuf**, S. Chatterjee, and M. Rasool, "COVID-19 Outbreak Cases and Healthcare Database," in *2021 International Conference on Intelligent Engineering and Management (ICIEM)*, IEEE, 2021. DOI: 10.1109/ICIEM51511.2021.9445298. (Scopus)
2. S. Hasan, **M. M. Yousuf**, M. Farooq, N. Marwah, S. A. A. Andrabi, and H. Kumar, "e-Vaccine: An Immunization App," in *2021 2nd International Conference on Intelligent Engineering and Management (ICIEM)*, IEEE, 2021. DOI: 10.1109/ICIEM51511.2021.9445386. (Scopus)
3. **M. M. Yousuf** and N. Saqib, "Design and Implementation of Shortest Path Line Follower Autonomous Rover Using Decision Making Algorithms," in *IEEE-Sponsored Asian Conference on Innovation in Technology (ASIANCON)*, 2021. (Scopus)
4. N. Saqib, **M. M. Yousuf**, and M. Rashid, "Design and Implementation Issues in Autonomous Vehicles – A Comparative Review," in *2021 2nd International Conference on Computation, Automation and Knowledge Management (ICCAKM)*, Dubai, UAE, IEEE, 2021, pp. 157–162. DOI: 10.1109/ICCAKM50778.2021.9357706. (Scopus)
5. M. Rashid, A. Hamid, N. Ahmad, M. B. Rehman, and **M. M. Yousuf**, "Novel Machine Learning Approach for Sentiment Analysis of Real Time Twitter Data with Apache Flume," in *2020 Sixth International Conference on Parallel, Distributed and Grid Computing (PDGC)*, IEEE, 2020, pp. 336–340. DOI: 10.1109/PDGC50313.2020.9315782.
6. N. S. Vishnu, **M. M. Yousuf**, S. M. Lakshmi, and M. Rashid, "MANET Congestion Alerting and Adapting Protocols – A Comparative Study," in *2020 International Conference on Intelligent Engineering and Management (ICIEM)*, IEEE, 2020, pp. 228–233. (Scopus)
7. M. Rashid, **M. M. Yousuf**, B. Ram, and V. Goyal, "Novel Big Data Approach for Drug Prediction in Health Care Systems," in *2019 International Conference on Automation, Computational and Technology Management (ICACTM)*, IEEE, 2019, pp. 325–329. (Scopus)
8. I. Nazeer, **M. M. Yousuf**, M. Rashid, and U. I. Wani, "Medical Diagnosis Cloud Model for Detection of Kidney Diseases," in *2019 Fifth International Conference on Image Information Processing (ICIIP)*, IEEE, 2019, pp. 315–320. (Scopus)
9. **M. M. Yousuf**, "Reducing the Software Maintenance Cost – An Empirical Study on the Python Software Systems," in *ICAST-2018*, Rayat Bahra, Ropar, 2018. [Best Paper Award]
10. **M. M. Yousuf**, "A Survey on Cloud Computing – A Real Buzzword in the IT World," in *National Conference on Computer Engineering Problems Optimization (CEPO-2016)*, Central University of Punjab, Bathinda, 2016.
11. **M. M. Yousuf**, "Emergency Communication – An Android App," in *11th JK Science Congress*, University of Kashmir, Srinagar, 2015.

C. Book Chapters

1. **M. M. Yousuf**, S. A. Sofi, and F. A. Reegu, "Challenges of Cyber-Physical Systems in Rural Education," in *Cyber-Physical System for Rural Development*, pp. 186–200, 2025.
2. **M. M. Yousuf**, S. Khare, A. A. Gadoo, and M. Rashid, "Blockchain Security in Cloud Computing: Use Cases, Challenges and Solutions," Springer Nature (Scopus).